

Nahid Hasan Sagar

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PROFESSIONAL SUMMARY

M.S. candidate in Statistical Science with experience in Bayesian modeling, time-series research, regression analysis, statistical learning, and scientific computing. Develops and benchmarks research software in R, C++, and Python, including Bayesian neuroimaging methods that achieved a $4\text{--}5\times$ computational speedup. Combines rigorous graduate training with university teaching experience and nine years of quantitative instruction.

EDUCATION

Indiana University, College of Arts and Sciences

Master of Science in Statistical Science | GPA: 3.68/4.00

Bloomington, IN

Expected May 2027

Thesis: Time-Series Analysis Using Bayesian Additive Regression Trees (BART)

Graduate Coursework: Introduction to Statistical Computing; Applied Statistical Computing; Fundamentals of Statistics I–II; Applied Linear Models I–II

Current Coursework: Statistical Consulting; Research in Statistics; Introduction to Analysis I

University of Dhaka

Bachelor of Science in Physics

Dhaka, Bangladesh

August 2024

EXPERIENCE

Research Assistant, StatMIND Lab

Indiana University Bloomington

2025 – Present

Bloomington, IN

- Develop computational methods to improve the efficiency and scalability of Bayesian neuroimaging pipelines using R, and C++.
- Achieved a $4\text{--}5\times$ speedup in Bayesian neuroimaging algorithms and benchmarked performance using simulated and real fMRI data.
- Build reproducible benchmarking workflows to compare runtime, scalability, and computational performance across implementations.

Teaching Assistant, Department of Statistics

Indiana University Bloomington

2025 – Present

Bloomington, IN

- Support undergraduate Statistics for Life Sciences courses through classroom instruction, discussion sections, and individualized guidance.
- Evaluate approximately 200 assignments per grading cycle and provide detailed feedback on statistical reasoning, computation, and interpretation.
- Conduct weekly office hours and exam-review sessions to strengthen students' conceptual understanding and problem-solving skills.

Senior Mathematics Instructor

Udvash Academic & Admission Care

2016 – 2025

Dhaka, Bangladesh

- Taught and mentored more than 10,000 students through classroom instruction and national online learning platforms.
- Prepared students for competitive university admission examinations through structured lessons, practice materials, and academic planning.
- Translated advanced mathematical concepts into accessible problem-solving strategies for students with varied educational backgrounds.

SELECTED RESEARCH & PROJECTS

M.S. Thesis: Time-Series Analysis Using BART

R, Bayesian Modeling

- Conduct thesis research on the use of Bayesian Additive Regression Trees for time-series analysis under the supervision of Dr. Pratola.

Ordinary Least Squares from Scratch

Python, NumPy, GitHub

- Develop an educational implementation of ordinary least squares regression from first principles, emphasizing matrix-based estimation and transparent statistical computation.

TECHNICAL SKILLS

Statistical Methods: Bayesian inference, BART, MCMC, hierarchical modeling, time-series analysis, linear and logistic regression, mixed-effects models, bootstrap methods, experimental design, and model selection

Statistical Learning: Ridge regression, lasso, decision trees, random forests, boosting, cross-validation, classification metrics, and predictive model evaluation

Programming: R, Python, C++, Rcpp

Computing and Tools: Numerical optimization, algorithm benchmarking, code profiling, unit testing, Git, GitHub, LaTeX, RStudio, and VS Code

Languages: Bengali (native), English (fluent), Hindi (conversational), Urdu (conversational)

LEADERSHIP & HONORS

- Attended the 2026 Velocity Conference at UC Berkeley as an Indiana University representative for the science community.
- Collaborated on the development of a comprehensive online Linear Algebra curriculum during the COVID-19 pandemic.
- Co-organized the Australian Mathematics Competition in Bangladesh in 2014.